James Watt and the Law of Patents

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I

Historians once more begin to stress the importance of the law in facilitating or obstructing economic growth.[1] As far as the patent law is concerned, there has been no lack of comment by historians of technology and of industrialization about its effects. Indeed, there is hardly one who has not felt it incumbent to express some opinion about the matter, and yet there has been little solid investigation of the patent law in the second half of the 18th century by anyone except members of the Patent Office staff themselves.[2] Even Sir William Holdsworth's mammoth *History of English Law* helps us very little. The section on 18th-century patent law is very short, highly derivative from papers published early in the *Law Quarterly Review*, and often confusing;[3] yet it manages to reach the firm conclusion

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[1]R. M. Hartwell, at the Conference of Economic Historians in Atlantic City, N.J., in September 1971, took the historians to task for failing to interest themselves in those aspects of legal history which had a direct bearing on problems of economic growth.

[2]E.g., R. M. Hartwell, *The Causes of the Industrial Revolution* (London, 1967), p. 39; M. W. Flinn, *Origins of the Industrial Revolution* (New York, 1966), pp. 70–71; T. S. Ashton, *The Industrial Revolution 1760–1830* (London, 1948), p. 12; P. Mathias, *The First Industrial Nation* (New York, 1970), p. 37. These books present a fair range of opinion about the effects of patent law. Witt Bowden, *Industrial Society towards the End of the Eighteenth Century* (1925; 2d ed., London, 1965), shows a rather deeper investigation of the problem of the English patent law during the Industrial Revolution than has been undertaken by more recent economic historians but is far from satisfactory. A. A. Gomme, *Patents of Invention*, British Council Pamphlet (1946), and H. Harding, *Patent Office Centenary*, H.M.S.O. (1953), are two useful pamphlets written by members of the staff of the Patent Office. They are concerned with the state of the law at different periods rather than with the economic implications of the law.

[3]William Holdsworth, *History of English Law* (London, 1938), 11:425–38. The most important articles in the *Law Quarterly Review (L.Q.R.)* are: E. Wyndham Hulme, "On the Consideration of the Patent Grant, Past and Present," *L.Q.R.* 13 (1897): 313–18; "On the History of the Patent Law in the Seventeenth and Eighteenth Centuries," *L.Q.R.* 18 (1902): 280–88; "Privy Council Law and Practice of Letters Patent for Invention from the Restoration to 1794," *L.Q.R.* 23 (1907): 180–95; D. Seaborne-Davies, "Early History of the Patent Specification," *L.Q.R.* 50 (1934): 86–109.

that "during the latter part of the seventeenth and the eighteenth centuries the administration of the law as to the grant of patents, and those special provisions made from time to time by the Legislature, was successful in encouraging British industries."[4] Earlier periods of the history of the patent law have been well treated by J. W. Gordon *(Monopoly by Patent* [1897]), W. H. Price *(The English Patents of Monopoly* [1913]), and H. G. Fox *(Monopolies and Patents* [Toronto, 1947]); but just at the stage in history when the consequences of the patent law might be of crucial importance we begin to wallow in ignorance.

If recent historians are unable to help us greatly, perhaps the legal experts of the later 18th century—the emergent patent agents and the counsels and attorneys who were beginning to specialize in patent cases—may be able to help us from their nearer view. It is, however, problematic whether if we had all their papers available to us we should learn that much more, if we respect the opinion of Abraham Weston, one of Boulton and Watt's attorneys, who reported in the following terms to the Committee of Patentees in 1785:

> It might therefore, it would seem very reasonably to be expected that the Law Books, which in general furnish very full statements of the modern Law should furnish much light and information in the matter of Patents. The fact is however otherwise, for whether it has happened that Questions between Patentees, or about Patents have commonly been Questions of fact and not of law, which I take to be the Case, or that the general Questions of Law on the Subject have never been brought forward on any important Trial, or from whatever cause it has arisen, it may with truth be said that the books are silent on the subject and furnish no clue to go by, in agitating the Question "What is the Law of Patents?" In the reports since Lord Mansfield has sat on the bench, there are not even the Titles "Patent" or "Monopoly" in the Indexes to any of the reports of Cases adjudged in his time; tho' it is very well known, that a great number of Patent Cases have been tried before him; Nor are there any other of the Books that furnish any information on this head.[5]

In a similar but shorter vein, Chief Justice Eyre, in *Boulton and Watt* v. *Bull* (1795), found that "patent rights are nowhere exactly described in our books."[6] That lack of certainty was still there in 1829

[4]Holdsworth, 11:432.

[5]Observations of Patents (J. Weston supplied), Parcel *E*, Boulton and Watt Collection, Birmingham Reference Library.

[6]Holdsworth, 11:425.

when the Select Committee on Patents reported,[7] and it prompted the *Quarterly Review* to a savage denunciation of the system:

> All nations however low in the scale of civilization have agreed in protecting the arts of industry, and the liberality of our ancestors devised a scheme for the same purpose. This scheme constitutes the patent laws of Great Britain,—a system of vicious and fraudulent legislation, which, while it creates a factitious privilege of little value, deprives its possessor of his natural rights to the fruit of his genius, and which places the most exalted officers of the state in the position of a legalized banditti, who stab the inventor through the folds of an act of parliament and rifle him in the presence of the Lord Chief Justice of England.[8]

This view was shared by many who gave evidence before the Select Committee.[9] Throughout the evidence runs the same theme—the uncertainty of the law and the unpredictability of the judges' decisions. Yet this is the system of law which, according to some, stimulated innovation by guaranteeing security to the patentee, and which attracted foreign inventors to England for the same reason.

II

If the law books are deficient in the evidence they provide of the state of the patent law in the second half of the 18th century, there is one source of material which is particularly rich—the private and business papers of Matthew Boulton and James Watt, for the most part assembled in three main collections: the Boulton and Watt Collection in the Birmingham Reference Library, the Boulton papers known as the Tew Manuscripts in the Assay Office Library, Birmingham, and the family papers of the Rt. Hon. David Gibson-Watt, M.C., M.P., at Doldowlod in Radnorshire.[10] It is natural

[7]*Report from the Select Committee on the Law Relative to Patents for Inventions* (1829).

[8]Charles Babbage, "Reflexions on the Decline of Science in England, and on Some of Its Causes," *Quarterly Review* 43, no. 86 (1830): 305–42.

[9]*Report from the Select Committee* (1829): evidence of Mark Isambard Brunel, p. 39; Francis Abbot, p. 63; William Newton, p. 72—"there being no existing basis of law, the dictum of the judge is one thing one day and another thing another"; Joseph Merry, p. 88; Samuel Clegg, p. 95; Walter Henry Wyath, p. 104; Benjamin Rotch, p. 108; et al.

[10]I am indebted once more to Major Gibson-Watt for permission to quote from his papers, and also to Mr. Arthur Westwood, the assay master, Birmingham, and to the chief librarian, Birmingham Public Library. Some of the principal documents cited in this paper may be found in Eric Robinson and A. E. Musson, *James Watt and the Steam Revolution* (New York, 1969).

that there should be many relevant papers in these collections when one recalls that Watt, the greatest of British inventors, was closely involved with the patent law from 1769 until the expiry of the term granted by the Fire Engine Act (1775) in 1800, and that his partner, Boulton, besides having a large financial stake in Watt's patents, had several other patents in his own name. In addition the law cases in which Boulton and Watt were concerned in order to defend the major Watt patent raised most of the important issues of the law during the last quarter of the 18th century. Finally, and this is not generally recognized, Watt was probably the best-informed authority of his time on the patent law and personally prepared many of the documents in his own cases. He was also the most imaginative and creative source of ideas about the ways in which the law of patents could have been reformed, and his proposals anticipate and even extend beyond most of the reforms that have taken place in the English law of patents until the present day.[11] The attorneys, counsel, and witnesses on both sides in the legal battles, the judges, and Boulton and Watt's assistant, John Southern, also contribute largely to our information on the topic, for it is clear that certain specialists in the patent law were beginning to emerge by the last quarter of the 18th century if not before.

III

The central criticism of the patent law by patentees was that it became next to impossible to specify an invention in a way that would satisfy the courts, and this because it had never been laid down exactly what a specification should do. There is no mention of the specification in the Act against Monopolies (1624), and the custom arose in the first place merely to distinguish one invention from another. It was not the means by which the patentee disclosed the details of his invention to the public, since in the earlier history of

[11]The main sources of information about Watt's proposals for reform of the patent law are two versions of his "Thoughts upon Patents for exclusive Privileges for new Inventions" and two versions of his "Heads of a Bill to explain and amend the Laws relative to Letters Patent and grants of privilege for new Inventions." The more finished draft of the former document is located in Box 21, Boulton and Watt Collection, Birmingham Reference Library (and is hereafter referred to in footnotes as "Thoughts" *A*). The less finished draft of the first document is to be found in the same collection in Parcel *D*. Miscellaneous (and is hereafter referred to in footnotes as "Thoughts" *B*). The more finished version of the second document is in the Boulton and Watt Collection, Box 25 (and is hereafter referred to in footnotes as "Heads of a Bill" *A*). The less finished draft of the second document is at Doldowlod (and is hereafter referred to in footnotes as "Heads of a Bill" *B*).

the patent law it was thought that the patentee fulfilled this obligation by practicing his invention within the realm. Specifications were first customarily required to be filed about 1734.[12] The attribution of a special function to the specification is usually said to have been the responsibility of Lord Mansfield in *Liardet* v. *Johnson* (1778), when "the doctrine of the instruction of the public by means of the personal effects and supervision of the grantee was definitely and finally laid aside in favour of the novel theory that this function belongs to the patent specification."[13] Once the specification had to bear all this weight, all sorts of argument were brought against particular specifications—that they were not sufficiently clear or full, that they were not accompanied by drawings or models, that they confused something new with something already well known. When James Watt sought his first patent, he and his friends clearly regarded the specification as a very serious matter, but it is unlikely that they thought that it would have to bear the entire weight of making the invention known to the public.

Several drafts of Watt's first specification in his own hand exist among the papers at Doldowlod, but they are undated and it is extremely difficult to place them in order. In November 1768, Dr. John Roebuck, Watt's partner at the time, wrote: "I am not so very desirous of hastening the Patent. I am afraid we shall be obliged to specify before it is in our interest."[14] On the other hand, Matthew Boulton and William Small in Birmingham were urging Watt on lest there should be a premature disclosure of the invention or someone should get a patent first for the same invention. A most important letter on the subject of the specification was written by Small to Watt on February 5, 1769:[15]

> Mr. Boulton and I have considered your paper, and think you should neither give drawings nor descriptions of any particular machinery, (if such omissions would be allowed at the office) but

[12]Seaborne-Davies, p. 90.

[13]Hulme, pp. 317-18. There seems little doubt that Lord Mansfield did make a decision of such a kind in this case, but Hulme points out that our knowledge of this case rests almost entirely upon a letter by the engineer, Joseph Bramah, who was present. What Hulme does not point out, however, is that the letter is a published document—*A Letter to the Rt. Hon. Sir James Eyre, Lord Chief Justice of the Common Pleas; on the subject of the cause, Boulton and Watt v. Hornblower and Maberley* . . . (1797)—and that in the Boulton and Watt case Bramah was a witness for their opponents, had been roughly handled by Boulton and Watt's counsel, and was himself guilty of infringing Watt's patent. The document is therefore not an entirely innocent one.

[14]Roebuck to Watt, November 9, 1768, Doldowlod.

[15]Doldowlod.

> specify in the clearest manner that you can, that you have discovered some principles, and thought of new applications of others, by means of both which joined together, you intend to construct steam engines of much greater powers, and applicable to a much greater number of useful purposes than any which hitherto have been constructed, that to effect each particular purpose you design to employ particular machinery, every species of which may be ranged in two classes. One class for producing reciprocal motions, and another for producing motions round axes.
>
> As to your principles, we think they should be enunciated (to use an hard word) as generally as possible, to secure you as effectually against piracy as the nature of your invention will allow.

For his advice in this letter, Small has been criticized by H. W. Dickinson on the grounds that he and Boulton "advised Watt badly over the specification and in consequence he made two serious errors of judgment, firstly in not appending drawings, although he had proposed them, and secondly in patenting a principle of action and not an application of a principle."[16]

As we have seen, there seems to have been no requirement nor even any expectation for many years after 1769 that a specification should have drawings accompanying it. Second, Watt's patent was for a "Method of diminishing the consumption of fuel in fire-engines." If the patent had been in the first place for a steam engine, it might have been more logical to accompany the specification with drawings; but Watt, rightly or wrongly, was trying to avoid taking out a patent for a particular engine, closely described and specified, because he anticipated that infringers would change his engine slightly and then claim that it was a different engine. Despite his foresight, this happened, as in the case of Edward Bull, who simply inverted the engine. Patenting a "method" was common in the 18th century, and there were many patents with the word "method" in the title. As for the "patenting a principle of action and not an application of a principle," such language was fraught with difficulty in the 18th century.

In 1784 Aimée Argand, who was in the process of taking out a patent for his lamp, told Boulton:

> I wrote my Specification myself and made it as general and comprehensive as I could, as you'll see by the copy I join to my letter. having not time enough to send it to you and wait for

[16]H. W. Dickinson, *James Watt, Craftsman and Engineer* (Cambridge, 1936), p. 52.

> your remarks about it, I consulted Mr Fearn celebrated Advocate in Fetter Lane and very well acquainted with the Specifications and the philosophical Subjects of them, our friend Mr. Moore [Samuel More, secretary of the Society of Arts], Mr. Lloyd F.R.S. and Lawyer. these Gentlemen found my Specification very good and made very few alterations to it. they all agreed in saying that there was no need of particular description and drawings, because the patent was taken upon the principle which may be applied to numberless shapes and forms, whereas giving particular description and drawings would be confining ourselves to these particular forms and enabling others to use the same principle under other forms. It is not the case of any composition of various machines or of any compound and complicated machine, therefore everyone seeing a Lamp and reading the Specification in order to understand fully the principle will be either in possibility of imitating the Lamp at the expiration of the patent, or in the impossibility of running his head against it by any contrivance of the same kind. . . . One thing grieves me only about the Patent, which is our non sense in having spoken in the petition. I was fool enough to let Mr. Parker write himself of a *new invented Lamp,* instead of having said a *new method of gaining light.*[17]

Boulton in reply complimented Argand on having "very properly described the spirit of the invention and avoided mechanical minutiae."[18]

Here are all the problems of Watt's 1769 specification all over again. The specification is in general terms; it avoids any particular description; it includes no drawings; the purpose is to avoid piracy being perpetrated by those who would make minor modifications; the specification is to be eked out by reference to the lamps themselves; and Argand would have preferred to take out a patent for a method rather than a lamp. Throughout his letter Argand talks about "principle." Argand had consulted with Boulton and Watt but also with lawyers and other specialists in patents in London who agreed that he was doing the right thing, and all this six years after the case of *Liardet* v. *Johnson,* which had clearly done very little to clarify the minds of the lawyers about the exclusive weight now to be laid upon specifications. As soon as Argand's patent was taken out, infringements began, but when the matter was taken to court the patent was defeated largely by the evidence of J. H. de Magellan, who testified that a similar lamp had entered England from France a

[17]A. Argand to M. Boulton, July 4, 1784, Assay Office Library, Birmingham.
[18]M. Boulton to A. Argand, July 31, 1784, Assay Office Library, Birmingham.

few months before Argand took out his patent. Had Argand's case not been settled upon this point, it is possible that all the objections brought against Watt's specification might have been brought before the courts ten years earlier.[19] Argand's fears and Boulton and Watt's fears were the same—specify too particularly and the patent would be evaded. No wonder that Boulton wrote to Watt with the suggestion: "Be it therefore enacted that if any person take out a patent for another persons inventions although he change the names and forms of the parts of Machines invented by others in order to evade the patent right of such other inventions and though some other invention may be combined therewith, yet such patents shall be void to all intents and purposes and the first and true inventor shall enjoy the advantages of his own inventions without being infringed upon by any subsequent inventors."[20]

Naturally Boulton and Watt's opponents made maximum play with the deficiencies in Watt's specification, and though they raised other shortcomings, this question of "principle" was the most serious. Bull's counsel, Sergeant Le Blanc, contended in 1794 in *Boulton and Watt* v. *Bull* in the Court of Common Pleas: "The reason seems obvious why this privilege of a monopoly which is to be granted by the Crown should not be granted merely for the Principle or for the first idea which may occur to an ingenious mind because if that is the case he is to reserve to himself the sole power of every possible improvement which may be made upon that idea in bringing it forward to perfection in the shape of a complete instrument."[21] When the judges came to their decisions, they divided two and two. Mr. Justice Rooke seems to have had the clearest understanding of the issues involved. He pointed out that the principles of natural forces were not involved in Watt's specification. Clearly Watt was not trying to patent the expansive force of steam.[22]

[19]One good thing for Boulton and Watt came of Argand's defeat. The leading counsel opposing Argand was Thomas Erskine, and Boulton and Watt, learning from Argand's misfortune, took care to have him on their side.

[20]M. Boulton to J. Watt, September 19, 1784, James Watt Box 3, 1782-84, Assay Office Library, Birmingham.

[21]"Boulton and another versus Bull: Copy from Mr. Gurney's Short Hand Notes of the Argument in the Court of Common Pleas 28th June 1794: Mr. Sergeant Le Blanc's Argument for the Defendant: 1st Argument, p. 12," Doldowlod.

[22]Watt himself, in his "Thoughts" *B*, had difficulties in defining what he meant by principles. In discussing the "principles" employed in the Marquis of Worcester's engine, Savary's engine, and Newcomen's engine, he saw the principle employed in the first engine as "simply *the force of steam*. the second added to the force of steam, the power gained by *condensing the steam with cold water*, but in both those the steam was

Whether or not the steam was condensed separately from the cylinder, the expansive force of steam still remained as a principle in both a Newcomen engine or a Watt engine. What Watt had done was to devise a mechanical arrangement whereby greater advantage might be taken of the principle of the expansive force of steam.[23] Lord Chief Justice Eyre agreed that the introduction of the word "principle" in the specification had introduced confusion. In his opinion: "Undoubtedly there can be no Patent for a mere Principle but for a Principle so far embodied and connected with corporeal Substances as to be in a Condition to act . . . I think there may be a Patent for."[24] Watt had therefore rightly been given a patent on that basis. As for Mr. Justice Heath and Mr. Justice Buller, they seem not only to have been confused about what they meant by "principle" and to have equated it with "method," but they went on to deny that a patent could be granted for a method, though hundreds of such patents had in fact been granted and their view would have totally disrupted the whole patent system. Mr. Justice Buller even contended that one could not have a patent for a chemical method of producing something more cheaply or efficaciously, but that one had to produce a new substance. Such a view would have blighted all attempts to secure patents for manufacturing vitriol in lead chambers or producing alkali from common salt.

These issues were still in debate in 1829, although it seems to have been the consensus that patents could "not be granted on abstract principles." The examination of John Taylor shows that the position was still far from clear. Taylor thought it inconvenient that good principles should be lost because a "person may not immediately have a technical mode of applying such principles."[25] Patentees, to get over the difficulty of the laws, were in the habit of describing not

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admitted to come into contact with the water to be raised. The third employed a weaker steam just sufficient to displace the air in the Cylinder and counter ballance the pressure of the atmosphere on its piston. His new principle was the *action of the pressure of the atmosphere on a piston sliding in a Cylinder* instead of causing the steam to act immediately on the surface of the Water to be raised and the making the machine *work pumps*." Though Watt says in this section of his draft that "by Principle is meant the general outline of the manner of producing the desired effect," his examples, particularly in the Marquis of Worcester's engine and in Savary's engine, are perilously close to a description of simple natural forces. In Watt's "Thoughts" *A* this section was omitted.

[23] *The Special Case in the Cause Boulton and Watt against Bull in the Court of Common Pleas* . . . (London, 1795), pp. 6–8.

[24] Ibid., p. 24.

[25] *Report from the Select Committee* (1829), p. 11.

only a principle but as many different methods as they could think of for applying it, with the result that the specification became complex and cumbersome. Davies Gilbert thought that the discovery of a principle was far more important than any application.[26] John Farey thought that "if the patent were given for the principle exclusively, it would be pernicious to the public, that other inventors should not be permitted to work upon that principle by other methods of execution, so as to produce a better result."[27] He thought that at first the inventor should be given a monopoly of all applications of his principle by his preliminary specification and then in his full specification choose the application which he thought most effective. Arthur Aikin's evidence specifically dealt with Watt's invention:

> Supposing a new principle to be discovered and applied to practice in one efficient mode by the inventor, should he be entitled to a patent protecting the principle, however other applied?—That I think would depend upon the nature and extent of the principle. Take, for example, the principle of condensing steam in a separate vessel?—I should be inclined to allow Mr. Watt, the inventor, to have a monopoly of it, provided in his specification he had given (which he has not given) all the details of the machinery employed by him at the time of taking out his patent, whereby he carried the principle into practice.[28]

Albert Roche's evidence is a lengthy discussion of the complete failure of the judges to decide among themselves what was a principle and what a principle reduced to practice.[29] He also believed that an inventor should not be required to enumerate all the applications of his principle and that consequently the only patent worth having was one for a new principle. He refers to an invention for securing a greater speed of distillation by throwing a wash into the boilers in the form of a shower so that a greater surface is exposed to the heat. He points out that the shower effect might be secured by a thousand different ways and that the inventor cannot be expected to enumerate them all, and that he deserves protection from another man who simply devises a different way from those enumerated for obtaining the same effect. Thus the basic problem of Watt's specification was still not resolved sixty years later. It is perhaps ironic that if the Fire Engine Act of 1775 had not stated that any objection to the patent of 1769 would be an objection to the Act

[26] Ibid., p. 15.
[27] Ibid., p. 35.
[28] Ibid., p. 45.
[29] Ibid., pp. 106–18.

itself, Watt would have been home and dry, because the Act referred to "engines" rather than to "methods" and for the Act Watt supplied drawings.

IV

Turning his mind to ways of clarifying the situation, Watt drafted several suggestions relevant to this central problem. The more discursive work is Watt's "Thoughts upon Patents . . . ," [30] in which he points out that there is no statutory requirement of the patentee to specify, but that the requirement rests solely on the proviso in the patent; that the requirement "seems to have been originally intended not so much as to secure the public in the secret of the invention, *as to discriminate one inventor's property from that of another*"; and that the proviso in the patent does not require the patentee to specify in such a way that from the specification alone his invention shall be made known to the public. In all these matters Watt has been shown to be historically accurate. He then goes on to suggest that the patent should be obtained "in the ordinary way" and that after four calendar months the patentee should be required "to deliver in a general specification of the *principles* on which his invention proceeds, and that in the most comprehensive short and clear manner which the patentee can devise." Within twelve months after the date of the patent, the inventor "shall deliver in a particular description of his invention, explaining at least one way of putting his principles in practise." This second specification should be accompanied by drawings where necessary. The inventor may now suggest what other applications might be made of his invention, but he is not to be expected to list them all, so that any application, mentioned by the patentee or not, is covered by his patent.

In the draft at Doldowlod of "Heads of a Bill for explaining and amending the Laws relative to Patents,"[31] Watt sets out a long section in which he tries to protect the patentee from infringers who make minor alterations in the forms and shapes of the invention and thereby seek to evade the patent. But he also specifically declares:

> that nothing herein contained shall be so construed as to authorize any patentee or holder of exclusive privilege to prevent any person whomsoever from using any Principle, Method art process ingredient, instrument or utensil which has been in common use before the Date of his Application for his Patent or from applying the same to any use or purpose whatsoever

[30]Boulton and Watt Collection (see n. 11 above).
[31]"Heads of a Bill" *B*.

> excepting always to such new uses and applications thereof as shall be of the patentees invention and set forth as such in his General or particular Specification, or in the title to his patent, nor to prevent any person whomsoever from making machines instruments utensils or commodities to answer the same purposes or to produce the same effects, by any principles methods arts or processes which shall not be of the Patentees invention or not set forth in his specifications, or not deducible from or parts of these contained in the said Specification.

He also includes the statement: "No patent to be valid for the sole use of any natural substances but for particular applications or uses of the same, *such being new*."

At the same time Watt rejects the Mansfield doctrine that it is from the specification alone that the public shall be informed how to make use of the invention at the expiry of the patent. For example, in his "Thoughts upon Patents" he declares: "No patent now in being to be set aside for a defective specification provided that the inventor shall prove that he has publickly practised the invention and that he has by himself or others instructed 5 or more persons, not bound to secrecy after the term of the patent in the use of, and method of constructing making or practising the same, so that the public are effectually secured in the possession of the invention."[32] The number of persons instructed is increased to ten in the Doldowlod draft of "Heads of a Bill," but the proviso is basically the same.[33]

Another part of the basic doctrine of the patent law ascribed to Lord Mansfield was the idea that the patentee must instruct in his specification not any member of the public but only persons skilled in that particular art. Thus the *Morning Post* for February 23, 1778 reported Mansfield as saying, "And the last thing is whether the specification don't teach any other artist to make use of it." Watt accepted this criterion in his various proposals, but it seems unlikely that the idea originated with Mansfield, since, on February 5, 1769, Small wrote to Watt in very similar terms: "I am certain that from such a specification as I have written any skilful mechanic may make your engines, altho it wants corrections, and you are certainly not obliged to teach any blockhead in the nation to construct masterly engines."[34] By 1795, Sergeant Williams, Bull's counsel, was stating the doctrine as a generally received criterion, not specifically emanating from Mansfield's judgment: "the patentee must describe his

[32]"Thoughts" *A*, fol. 34.
[33]"Heads of a Bill" *B*.
[34]Doldowlod.

invention in such a way that others in the trade who are Artists may learn to do the thing for which the patent is granted following the directions of the Specification,"[35] though he and his colleagues, unlike Watt, would not allow that the public working of the invention had any part to play in the instruction of these artists and insisted on the role of the specification alone.

Though Watt and his opponents did not agree upon the legal necessity for a specification nor upon the purposes that it ought to serve if there were to be one, the time of the courts was largely taken up in hearing the evidence of expert witnesses as to whether Watt's specification was sufficient to enable a person skilled in the art to erect an engine as good as one built by Boulton and Watt. In the Bull case, J. A. de Luc, William Herschel, Dr. James Lind, John Southern, Robert Mylne, Alexander Cumming, William Murdock, John Rennie, Jesse Ramsden, and Richard Mitchell all declared for the plaintiffs, Boulton and Watt, that they could have erected Watt's engine from his specification alone, while James Wood, Joseph Bramah, Jabez Hornblower, John Braithwaite, William Braithwaite, Thomas Rowntree, and Richard Trevithick, for the defendants, all gave evidence to the contrary. The scientists, de Luc, Herschel, and Lind, may be thought to have had little experience of the actual practice of constructing engines; Robert Mylne, as engineer to the New River Water Company, had had some experience; Ramsden and Cumming were makers of scientific instruments; Rennie, Mitchell, Murdock, and Southern were all in Boulton and Watt's employ. The numerical proportion of engine erectors was greater among Bull's witnesses, though they too were connected by interest with those for whom they gave evidence. The jury, however, without deliberating, came to the conclusion that Watt's specification was sufficient to enable an artist skilled in erecting steam engines to make Watt's engine. Though the question was again raised in later trials, the jury's conclusion on this point was of great advantage to Boulton and Watt. Nothing much is to be gained at this distance in time in debating whether the jury was right, but the contentions over this matter raised some important issues.

First of all, Watt recognized that if the technical questions about the specification could be removed at an earlier stage much time could later be saved. He therefore proposed in his "Thoughts upon Patents" an elaborate system for checking the specification at the outset. The inventor's second and more detailed specification was to

[35]"Mr. Gurney's short hand notes of the Case of Boulton and Watt versus Bull in the Court of Common Pleas 3 February 1795," Doldowlod, fol. 21.

be submitted to a committee of five persons—three members of the Royal Society, recommended by the Council of the Royal Society, together with two artisans skilled in the relevant trade, selected by the Master of Rolls from a list of ten supplied by the patentee. This committee would have the power to make any investigations it thought fit, call for models or drawings, interview workmen, etc. If it satisfied itself that the invention had been sufficiently differentiated from others and sufficiently explained, it would certify this to the Master of the Rolls, but if it was not satisfied it could direct the patentee to make out a better specification. After the commissioners had certified that the patentee had specified clearly, the specification would no longer be liable to challenge on that issue in any court of law. Commissioners would similarly advise on caveats to the taking out of patents. In the Doldowlod draft of "Heads of a Bill,"[36] the members of the Royal Society are retained, but in the Birmingham Reference Library draft they are dropped.[37] It should be observed that the commissioners were not to give their opinion on the utility or the novelty of the invention but were simply proposed as a way of assisting the patentee to draw up a specification that would instruct others how to carry the invention into practice. Watt's proposals here were farsighted, more thorough than the proposals for examiners accepted in the 19th century, and, it might be argued, in other ways superior to modern British practice.

Sir Eric Roll pointed out some years ago that the draft of Watt's "Heads of a Bill" in the Birmingham Reference Library contained annotations in the hand of Arkwright.[38] The same appears to be true of the more finished draft of Watt's "Thoughts upon Patents,"[39] though this has not been previously noticed. The second document also contains the argument that "if the Capital and meritorious invention such as Sir Richard Arkwrights cotton machine shou'd be brought forth in a Century in consequence of them [patents] it shoud justify the measure of granting them."[40] At their first acquaintance Watt and Arkwright were not drawn to each other, and there is no doubt that Boulton and Watt regarded Arkwright's behavior on various occasions as high-handed; but their common interest in protecting their patents, together with a mutual friend-

[36]"Heads of a Bill" *B*.

[37]"Heads of a Bill" *A*.

[38]Sir Eric Roll, *An Early Experiment in Industrial Organization* (1930; reprint ed., New York, 1968), p. 146, n. 2.

[39]"Thoughts" *A,* versos of fols. 18, 26.

[40]Ibid., fol. 42.

ship with Josiah Wedgwood, brought them together. In 1781, after Arkwright's defeat in the Court of King's Bench when he brought an action against Colonel Mordaunt for infringing his carding patent, Watt wrote to Boulton: "I fear for our own"; "I fear we shall be served with the same sauce *for the good of the public!*"[41] Consequently, in 1785 Watt gave evidence for Arkwright in two cases in which Arkwright tried to protect his carding patent. Watt recorded in his diary for 1785:

> Thursday 17 February 1785. Mr. Arkwrights cause tried before Lord Loughborough given in his favour with 1/- costs after a hearing of 9 hours.
> Sunday 12 June 1785. Sett out for London on patent cause before the house of lords and Arkwrights trial.
> Saturday 25 June 1785. Arkwrights Trial, Lost.
> Saturday 2 July 1785. Received from Arkwright/Draft for my trouble in his case £52.10–[42]

Wedgwood, reporting to Watt Arkwright's resentment of the treatment that he had received in the second trial, said that he would ask Arkwright to call upon Watt in Birmingham a few days after September 17, 1785 with the intention that the two should consult about the patent law.[43] It was presumably, therefore, at that time that Arkwright made his comments upon Watt's proposals. Arkwright's comments are not very substantial. In "Heads of a Bill" he merely suggests that the persons skilled in the appropriate art who are to be consulted about the specification should be referred to as witnesses rather than as commissioners, and that they should be nominated by the patentee rather than by the attorney general.[44] In the same document Arkwright suggests that a patentee seeking to add to his specification should petition the attorney general rather than the Lord Chancellor.[45] His comments upon Watt's "Thoughts upon Patents" amount to the suggestion that the general patent specifications should be printed annually by the King's Printer, a suggestion that is rather contrary to Watt's intention to keep even the general specification available only in the Chancery Records,[46] and a further

[41]S. Smiles, *Lives of Boulton and Watt* (1865), p. 303, cited by R. S. Fitton and A. P. Wadsworth, *The Strutts and the Arkwrights* (Manchester, 1958), p. 87, n. 3.

[42]Doldowlod.

[43]Wedgwood to Watt, September 17, 1785, Boulton and Watt Collection, Box 36, Birmingham Reference Library.

[44]"Heads of a Bill" *A,* verso of fol. 8.

[45]Ibid., verso of fol. 11.

[46]"Thoughts" *A,* verso of fol. 18.

suggestion on the same subject that would allow the detailed specification to be printed at the expiry of the patent.[47] In these last two suggestions Arkwright is seen to be rather more liberal than Watt.

V

If problems of specification were paramount in Watt's mind, he was also, like other patentees, very worried about the way in which specifications lay open to the gaze of anyone who was prepared to pay a small fee. Britain's fear that its industrial secrets were being stolen by its competitors found expression in the Act of 1785 forbidding the export of certain types of machinery and the emigration of skilled artisans.[48] In the absence of any international law of patents, it was all too easy for a foreigner to pick up the details of a new invention from its specification and then develop it to his own profit in his own country.[49] The problem did not arise, however, solely from foreign spies. There were many individuals and groups in Britain who were quite as ready as any foreigner to take an unfair advantage. In the uncertain state of the law, an attempt by a patentee to sue an infringer was fraught with peril, but the law compelled the patentee to expose his secrets without guaranteeing him any security. Moreover, during the period between his application for his patent and his submission of his specification, he was unprotected, and a competitor who got to know of his invention was potentially in a position to anticipate him in his own invention or to create evidence of a prior public disclosure.

Watt proposed in "Thoughts upon Patents" that an inventor should be allowed to enter a caveat with the law officers of the Crown against any person surreptitiously taking a patent for the same invention as his own.[50] There was also to be machinery for settling disputes between inventors who put forward similar inventions for a patent, but once a patent had been granted only the general specification was to be published. Particularized specifications were not to be shown to other persons except with the express mandate of the Lord Chancellor. There is no doubt that

[47]Ibid., verso of fol. 26.

[48]1785: 25 Geo. III, c. 27 prohibited export of certain machines and tools in the metal industries. 1786: a new act, 26 Geo. III, c. 89, contained a detailed list of tools and machines to be protected. See Witt-Bowden, (n. 2 above), p. 131.

[49]I am presently preparing a book on industrial espionage in this period in which I intend to deal with this subject at greater length.

[50]"Thoughts" *A*, fol. 17.

Watt's proposals, if they had been accepted, would have stopped a good deal of leakage of industrial technology, if only for a limited period of time. The more precise the requirements of specification became, the more complete the information available became to anyone who was able to consult the specification. In 1785 a Committee of Patentees was formed which objected strongly to the ease with which specifications were consulted,[51] and representations were made to the Lord Chancellor from that time forward, since, as Boulton wrote to him in August 1790: "From all I have been able to learn respecting the inrollment of specifications of Inventions I conceive that the controleing and disposeing power over them is entirely lodged in the Will of your Lordship."[52] But the fees received at the various offices were presumably sufficient to prevent any reform from taking place.

VI

There is in the Boulton and Watt collection a note in Watt's hand, undated, headed "Doubts and Queries upon Patents."[53] It dates probably from about 1795 and reads as follows:

> [1] Whether the King can grant a patent for a method of doing or performing any mechanical process
> [2] Whether in such case patent would be valid without a description of an *organized* machine
> [3] Whether a man improving his invention after patent granted, does not invalidate his patent
> [4] Whether a pattentee refusing to add his improvement to an old machine does not render patent void
> [5] Whether a patentee asking more than a common fair profit does not invalidate
> [6] Whether a patent for an improvement on an old invention is valid
> [7] Whether [a] patent for [a] new mode of using old instruments is valid
> [8] Whether a patent for chemical process is valid

None of these questions had been satisfactorily resolved by 1795. Queries 1, 2, and 8 may be considered together, and it can be seen that the accumulation of evidence submitted by Watt's counsel and attorneys in the cases against Bull and Hornblower and Maberley

[51] J. Watt to W. Matthews, July 20, 1785, Assay Office Library, Birmingham.

[52] M. Boulton to the Lord Chancellor, August 14, 1790, M. Boulton Box, Assay Office Library, Birmingham.

[53] Parcel *D*, Boulton and Watt Collection, Birmingham Reference Library.

finally convinced a majority of the judges that patents for mechanical and chemical processes had long been granted and that to reject them would totally disrupt the patent system.[54] As for query 2, this was a hare started by Lord Chief Justice Eyre in *Boulton and Watt* v. *Bull.* The question appears to have had no legal precedent, and Watt's counsel were able to show that any method for a mechanical process, if it were to operate effectively, had by its very nature to be organized.

Query 3, however, long remained unresolved. Consequently, in his "Heads of a Patent Bill,"[55] Watt included the following proposal:

> And whereas it doth and may so happen that patentees doth and may upon reflection discover some material defect in the explanatory part of their specification or hath made or may make some improvement upon their invention at some period after the same has been enrolled in Chancery in order to encourage the full disclosure of the inventions so secured or to be secured by patent as well as to secure the patentees in the possession of their rights. It is hereby enacted, that in any such case . . . every possessor of a patent . . . shall and may represent the same by petition to the Lord High Chancellor . . . and if it shall appear to the Lord High Chancellor that such alteration or addition to the said specification is material proper and essential to the said invention, he shall direct the matter to be referred to the Attorney General who shall cause the same to be examined by Commissioners and enrolled in Chancery.

This provision, however, was not to allow the patentee to embark on any new material or to prolong the life of his patent beyond fourteen years.

Watt's suggestion would appear to have been eminently sensible, since, as matters stood, if the patentee worked his invention in some way superior to his original specification, he was judged to have cheated the Crown by not making a full disclosure. In 1829 the matter was frequently discussed by the Select Committee,[56] and it

[54]E.g., Lord Chief Justice Eyre: "I believe I do not over-rate it when I state two-thirds, I believe I might say three-fourths of all Patents granted since the Statute passed, are for Methods of operating, producing no new Substance, and employing no New Machinery. If you look at the List [supplied by Watt's attorneys], I dare say you will find Fifty Patents for Methods of producing all the known Salts" ("Special Case in the Cause Boulton and Watt in the Court of Common Pleas," Boulton and Watt Collection, Birmingham Reference Library).

[55]"Heads of a Bill" *A*, fol. 11.

[56]*Report of the Select Committee* (1829), p. 11, John Taylor; pp. 27-28, John Farey; p. 42, Arthur Aikin; p. 47, Charles Few; p. 54, Benjamin Roche.

was agreed by all witnesses that the law operated to the discouragement of inventors who, obliged to specify before they had had time to carry out sufficient experiments, were then forced to keep their actual methods of working secret.

Query 4 was a problem of more special interest to Boulton and Watt, who were asked from time to time to add the separate condenser and air pump to an existing Newcomen engine and, of course, to charge a reduced royalty. Their evasions were many and they very seldom authorized such a conversion, but they were also careful not to give a direct refusal, since they felt that they might injure their patent by doing so:

> We still continue to be plagued with applications from persons having Common Engines for leave to adapt an Air pump to them. We have generally succeeded in dissuading them from the attempt but some of them return to the charge with so much positiveness that we are laid under the necessity of either asking a certain Sum, or of giving them a direct Negative. Were we to consult only our own inclinations, or our ultimate interest and the interest of the Parties without regard to legal inconvenience we should at once refuse them, but after the questions and difficulties which have been started upon this point, we are fearful of furnishing our Adversaries with a new ground of Complaint against us.
>
> You know we have in all our Arguments maintained that the Patent and Act of Parliament are merely for the application of our Principles (our Engines)[57] to the saving of Steam and fuel in fire Engines. If then, we refuse to apply or sell the right of applying those principles to common Engines; it appears upon the face of the matter, either that we consider them as inaplicable of themselves to common Engines, or that we claim the right of making the whole of the steam Engine, as now made by us. Certainly these are sufficient answers which may be given to these and other objections, but is it worth our while to incumber ourselves with new opposition when we may avoid it and pocket money into the bargain.[58]

The next query about fair profit was also a problem peculiar to Boulton and Watt, as far as we can judge, but it was potentially a

[57]Because the 1775 act was worded so that Watt's patent was continued for "engines" rather than for "principles," Boulton and Watt's lawyers were obliged to resort to the argument that the word "engine" could mean, in certain circumstances, "principle."

[58]James Watt, Jr., to A. Weston, November 4, 1796, Boulton and Watt Letter Book (Office), August 1795–July 1796, Boulton and Watt Collection, Birmingham Reference Library.

problem for any patentee who charged royalties. It is interesting to see the concept of the just price surviving so long in the business activities of the modern world.

When we turn to query 7, however, it is clear that this is a matter of central importance and immediately raises the question of what was meant by a "new manufacture" or "invention." In the harsh atmosphere of later 18th-century courts of law, patents were declared invalid if the patentee accidentally included in his specification as part of his own invention any part of a previously existing machine or process. Moreover it was sometimes argued that a patent for an addition or improvement, even if carefully distinguished from the preexisting machine or method, was invalid. Weston took the question up in 1795–96: "It appears to have been formerly held that an Addition to an Old Machine was not a New Manufacture within the Statute (3 Jas. I. 184) But this seems to have been over-ruled. In Morris v. Branson (Copy from Bullen's Nisi Prius p. 77) That a patent may be for an addition or improvement only appears to have been admitted in several Cases. Per Lord Mansfield in different cases, and by Buller J. in the King v. Elsie, Sittings at Westminster after Mic. 1785."[59] The judges in the Court of Common Pleas in 1795 clearly agreed that a patent for an improvement or addition was legal but found difficulty in deciding what was the nature of Watt's improvement or addition. For this reason, in his various proposals for reform of the law, Watt clearly stated that it would be legal to take out a patent for an improvement or addition, but he also added that if a patentee accidentally included within his claim something that was already known this mistake would not prejudice his claim to whatever in his patent *was* new. He also carefully stated what were to be understood by "new manufactures" and who would be entitled to a patent. The persons so entitled were:

> The Inventors of New Machines Arts and commodities
> The improvers of old Machines Arts or Commodities and of the method of construcing making or producing them
> Those who bring from Foreign Countries, the methods of making machines manufactures or commodities not practised or used in Britain before that time or at that time
> Those who combine together old Instruments or machines so as to produce new effects, or to make them more extensively useful to the publick

[59] A. Weston, "Remarks on Mr. Watt's Specification of his Method of Saving Steam and Fuel in Fire Engines," Doldowlod.

> Those who apply old machines or instruments to new uses . . .
> Those who by new processes produce common commodities in a better easier or more commodious manner.[60]

It will be seen that query 7 is covered in the fifth head above. Thus Watt makes provisions in his proposals to deal with all the queries raised in the list with which we began this section. Had his proposals for legislation been accepted they would not have continued to be matters for debate at least until 1852 and often until much later.

VII

The traditional picture of the relationships between Boulton and Watt shows Watt as an extreme hypochondriac, unskilled in business matters, and dependent upon his ebullient and self-confident partner, Boulton, for the reserves of confidence required in facing the obstacles and difficulties with which the partnership was confronted. Watt's dealings with patent law help us to revise the picture. Not only was Watt remarkably persistent in his defense of his patent, but he displayed that indefatigable power of analysis in his study of the patent law which characterized his genius in mechanical matters. Step by step he analyzes the problems, carefully assembling all the relevant evidence, and then moves to the exposition of his general principles. At each stage he defines his terms. From general principles he works through to the detailed point so that his proposals for legislation exhibit a closeness between the preamble and the succeeding clauses seldom bettered in English legislation.

Though Boulton and Watt hired the best lawyers available, there is no mistaking Watt's generalship. He says what is to be looked for,

[60]"Thoughts" *A*, fols. 15-16; cf. "Heads of a Bill" *A*, fol. 4. In the latter the words "any manner of new manufactures" are defined as:

1st every new art, machine, utensil, Manufacture, factitious substance or commodity

2d Every new improvement of or upon any Art machine Utensil manufacture substance or Commodity

3d Every new mode manner method or way of applying constructing or using any Art Machine utensil substance material or Commodity

4th Every new process for making preparing compounding or producing any manufacture, substance, material or commodity, whereby the same may be rendered more useful or valuable, may be made prepared or used in a more commodious or less costly manner, may be improved in its quality or otherways rendered cheaper more profitable or beneficial

Lastly and generally every new and useful Philosophical, Chemical or Mechanical Art or Invention whatsoever is hereby declared and enacted to have been and to be a subject properly intitled to be secured by patent, to the first and true inventors thereof . . .

he suggests the tactics, he assembles the witnesses and instructs them, and it is he with his letters and memoranda who tries to oblige his counsel to present a logical, well-argued case. Watt was probably the best patent lawyer of his time.

When we compare the attitudes of Boulton and Watt to the problem of defending the patent, it is Watt, not Boulton, who is determined to fight:

> Remember always that though the decree of a Judge should be against us which it cannot be, consistent with justice, that the house of peers remains, and I am told, after that the King in council but of the latter I am not certain. . . . However much I am vexed, I am ranged and shall prepare myself to meet the worst and not lie down to have my throat cut—I beg you would summon up your resolution and not lose the battle before you fight it—I pray heartily that your spirits may be supported and that you may be enabled to combat all your fears—If we are defeated in this matter we should be set up as a warning to all men how they put faith in patents or acts of a British parliament.[61]

Boulton himself is often much more diffident and uncertain: "To trial, or not to trial, that is the question. Whether 'tis better to sacrifice a part of our premiums which in all probability may never come into our pockets, than by grasping at all to loose all."[62] Watt had earlier chosen Ulysses rather than Hamlet for his model:

> I read some sentences in Shakespear lately which struck me as very applicable to our case
>
> Take the instant way;
> For honour travels in a strait so narrow
> Where one but goes abreast: keep then the path;
> For emulation hath a thousand sons
> That one by one pursue: if you give way,
> Or hedge aside from the direct forthright,
> Like to an enter'd tide they all rush by
> And leave you hindmost:
> Or, like a gallant horse fall'n in the first rank,
> Lie there for pavement to the abject rear,
> O'er run and trampled on: then what they do in present,
> Though less than yours in past, must o'ertop yours;

[61] J. Watt to M. Boulton, September 28, 1782, Boulton and Watt Letter Book (Office), April 30, 1782–July 20, 1784, Boulton and Watt Collection, Birmingham Reference Library.

[62] M. Boulton to A. Weston, ? 1795, Ambrose Weston Box 1, Assay Office Library.

For time is like a fashionable host
That slightly shakes his parting guest by the hand,
And with his arms outstretch'd, as he would fly,
Grasps in the comer: welcome ever smiles,
And farewell goes out sighing.[63]

After so much strength and determination, what was achieved? Boulton and Watt did prevail in their battle, but their situation was unusual. Arkwright was defeated, even with Watt as a witness; Argand lost his lamp patent; Wedgwood gave up patents as more trouble than they were worth. In general the attitude of the judges was unsympathetic to patents. Lord Chancellor Kenyon, even while giving his decision in favor of Boulton and Watt in the Court of King's Bench in 1799, said that "he was one of those who had not been a great favourer of Patents, because (though in many instances, and particularly in this instance, great benefits were derived from them to the country) from the balance on the whole, it struck him that there was a great deal of oppression of the lower orders of men from Patents, by those who were more opulent."[64] Watt, in a draft of his "Thoughts on Patents,"[65] had earlier argued against this prejudice that patents were "Encroachments on the rights of mankind in general" by asking: "How it can be considered as conferring a Monopoly to grant a man the exclusive privilege of using a thing which had perhaps never existed if his ingenuity or industry had not been exerted in discovering it and bringing it to perfection." It is clear, however, that many legal and political minds were doubtful of the advantages to the nation of a patent system, and many patentees were defeated as much by the unsympathetic attitude of the judges as by the ingenuity of their opponents' lawyers. Everywhere pirates triumphed, foreign spies sketched in their notebooks, key workmen absconded with their masters' secrets, judges contradicted each other and themselves while the attorneys grew fat and the patent agents emerged to take their share of the spoils. Yet in Watt's desk lay the basis of simple legislation which would have done much to clarify the situation. If adopted it would have led to greater security for the patentee, better encouragement for domestic and immigrant inventors, and at the same time discouragement of false specification and the taking out of patents for trivial matters. Nor was it that the

[63] J. Watt to M. Boulton, October 17, 1782, Boulton and Watt Letter Book (Office), April 30, 1782–July 20, 1784, Boulton and Watt Collection, Birmingham Reference Library. The quotation comes from *Troilus and Cressida*, act 3, sc. 2.

[64] *Times Law Report*, January 26, 1799.

[65] "Thoughts" *B*, fol. 3.

government was not informed of Watt's proposals or that there was no support in the country or in Parliament for reform of the patent law. But that is another story.

Boulton and Watt, after spending something like £10,000 to defend their patent, triumphed and thus remained as examples for the historian of the protective value of the patent law. According to John Farey, Watt was exceptionally fortunate. Asked if proper specifications would have secured Watt against any rivalry, Farey replied: "They did so in fact most completely for thirty years; his paper was most admirable well drawn and very definite; but in allowing those heads to pass for a complete specification, a latitude was given to Mr. Watt, in favour of his great services, *which the courts have never allowed in any other case*."[66] Farey continues by asserting that Watt never did make a complete disclosure: "the consequences of that omission have been important in his case, for long after the expiration of his patent . . . engineers who wanted to make steam engines, had to go and steal a knowledge of his invention from his factory, or from examining engines made by him, with as much difficulty as if he had never had a patent." Watt succeeded because of his great fame and because of the influence of Matthew Boulton, who during part of the critical period was high steward for the County of Stafford and hobnobbed with some of the judges who were deciding his case.

One's sense of reverence for the impartiality of the English judiciary at this period—even outside political cases, in which it has long been impugned—is severely shaken by the relationships between Boulton and Watt and the judges in these cases over the patent. Watt wrote a long letter to the chief justice of Common Pleas in March 1795,[67] which, though Watt says it is not about the case, is about exceedingly relevant matters. The lord chief justice himself goes a-gossiping with Mrs. Vere, the banker's wife, and tells her that if Boulton and Watt's patent "was to be tryed upon one point it would be found not to be worth a farthing."[68] Judge Rooke meets Boulton at Stafford Assizes and is evidently so charmed that when he returns to London he seeks out Boulton's attorney, Abraham Weston, and asks to be shown around the Watt engines at York Building

[66]*Report of Select Committee* (1829), p. 32. My italics.

[67]Boulton and Watt Letter Book (Office), September 1793–September 1795, fols. 163 ff., Boulton and Watt Collection, Birmingham Reference Library.

[68]M. Boulton to James Watt, May 13, 1797, Boulton and Watt Collection, Box 20, Bundle 18, Birmingham Reference Library.

and Shadwell.[69] Later that year Rooke and his wife were shown round Soho by Boulton's son—a visit in which, you may be sure, they were treated with every courtesy.[70] Boulton dines with Judge Buller in February 1795 but gets no change from him.[71] Of course the judges may not have been influenced by all this social conversation, but to modern eyes it would seem to have given Boulton and Watt considerable advantages over their opponents. The clubbishness of the 18th century was a very powerful spirit. A combination of Boulton's *savoir-faire* and Watt's persistence, backed by large sums of money, would have been difficult to resist in any age, but it was not a combination that the average patentee had at his disposal.

[69]M. Boulton to A. Weston, June 30, 1794, Boulton and Watt Letter Book (Office), September 1793-September 1795, Boulton and Watt Collection, Birmingham Reference Library.

[70]M. R. Boulton to M. Boulton, September 10, 1794, M. R. Boulton Box (1794-1842), Assay Office Library, Birmingham.

[71]James Watt, Jr., to M. R. Boulton, February 17, 1795, James Watt Junior Box 1 (1789-1810), Assay Office Library, Birmingham.